From Bystander to Participant: Design Factors Shaping the Participation Threshold in Curated Interactive Installations

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Abstract

An interactive installation's technical refinement means little if visitors do not engage with it. A recurring pattern across visitor studies is that people hesitate to participate even when they attentively watch an installation. This paper synthesizes findings from studies of interactive exhibitions and installations across curated and social settings to identify factors influencing visitors' decision to participate. Three overarching themes emerge across these studies: the legibility of an installation and its affordances, the "honeypot effect" of observing other visitors interact, the visibility and consequences of failure. Together, these factors shape whether an interaction appears approachable and socially safe. Building on these findings, the paper proposes design strategies that lower the participation threshold. Rather than treating hesitation as individual reluctance to engage, the framework situates participation within the social and spatial conditions created by installation design and highlights how design decisions can influence whether visitors remain observers or become participants.

1. Introduction

Interactive installations are a recurring feature of contemporary exhibitions, festivals, and public spaces, promising visitors the chance to influence and become part of the work instead of passive viewing. Yet in practice, many visitors remain bystanders: they might watch and sometimes read the label but move on without ever participating. This gap between observing and participating is not merely a matter of individual shyness or disinterest. It is shaped, often quite precisely, by decisions made in the design of the installation itself: how it is positioned in space, what it signals from a distance, and what it risks exposing about the person that interacts. This paper asks: what design factors determine whether a visitor crosses from observer to participant in an interactive installation, and how do these factors interact? Answering this question matters for two reasons. Practically, curators and installation designers routinely face this threshold as a design problem because an installation that fails to attract participants fails at its core purpose, regardless of its technical or artistic sophistication. Conceptually, the question sits at the intersection of several distinct research traditions: HCI, museum studies, and public interaction design that can be brought together for new findings. This paper's contribution is three-part:

- It maps the terminologically distinct role and threshold models used across three foundational studies onto a shared vocabulary, anchoring the analysis on the transition Wouters et al. term *audience member* → *participant*, which corresponds to Brignull & Rogers' *participation threshold* (Section 3).
- It synthesizes the findings from all three studies on recurring influencing factors and demonstrates that the social risk of failure is best understood as two distinct factors, namely, whether failure is structurally possible and whether it is visible to others rather than as a single dimension.
- It resolves an apparent contradiction between studies on uncertainty: Wouters et al. found that vague information is more effective in the earliest stage of awareness, while Brignull & Rogers found that explicit, forward-looking information is more effective closer to the participation threshold—this paper argues that these are effects specific to different phases of the same process, rather than contradictory findings (Section 3.3.3), and translates the resulting synthesis into concrete design implications (Section 4).

2. Related Work

The honeypot effect describes the observation that visible engagement attracts further engagement. It was introduced by Brignull and Rogers [BR03] in their study of the Opinionizer, a large public display onto which visitors could type opinions on a given prompt using an adjacent keyboard, with contributions appearing live alongside a cartoon avatar chosen by the participant. The system was deployed at two real social, rather informal gatherings instead of in a lab setting, allowing the authors to observe how the display was actually approached, avoided, and discussed within an ongoing social occasion. Their work remains foundational for this paper's framework, contributing the three-space model of peripheral awareness, focal awareness, and direct interaction, and offering some of the clearest evidence connecting social embarrassment

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to non-participation. Hornecker and Stifter [HS06] apply related ideas to a technology museum, combining smartcard tracking data, logfiles, and roughly 16 hours of observation with 30 visitor interviews to examine how visitor types of different ages engaged with a diverse set of exhibits, ranging from traditional object displays to computer-enhanced hands-on installations and screen-based media. Their study extends the honeypot effect into a curated, semi-public exhibition setting and provides detailed comparative data on engagement duration, group size, and interaction patterns across exhibit types. This paper draws on this data to compare exhibits with and without a defined failure state (Section 3.3.1), though this distinction is this paper's own framing rather than the original study's focus. More recently, Wouters et al. [WDH*16] propose the Honeypot Model from an in-the-wild study of Encounters, a large-scale interactive installation deployed in a public courtyard over a four-evening summer festival in Melbourne, in which overhead depth sensors tracked visitors' movements within three dedicated dance zones and translated them in real time into dynamic projected visuals, soundscapes, and lighting. Drawing on observation, interviews, and sensor-logged behavioral data, their model offers the most granular account to date, introducing a six-role taxonomy and identifying the honeypot sweet spot. Each study makes a substantial contribution: Embarrassment evidence (Brignull and Rogers), exhibit-level interaction comparative data (Hornecker and Stifter), and a fine-grained engagement model (Wouters et al.). These three studies have not previously been read alongside one another, despite their shared focus on the same underlying phenomenon across different public settings. This paper brings them into conversation by mapping their differing terminologies onto a shared vocabulary (Section 3) and drawing out a set of recurring design factors that cut across all three (Sections 3.1-3.3).

3. Mapping User Roles Across Studies

Wouters and Brignull both propose models that conceptualize visitor engagement as a progression through different stages of interaction. Wouters describes this progression through a series of visitor roles, going from passer-by (no participation) to dropout (finished participation). Within this model, an audience member is "familiar with the interactivity and social norms surrounding a system," while a participant "exhibits subtle forms of engagement," such as moving slowly within the interaction space [WDH*16]. The six-role model is relevant to this research because it explicitly conceptualizes the transition between different levels of engagement as a distinct step. Rather than treating engagement as a continuous progression, the model identifies specific changes in the visitor's relationship with an interactive system by connecting these roles to spatial factors. Brignull suggests a concept that works primarily spatially rather than primarily through visitor roles. Brignull distinguishes between three spaces: Space A, characterized by peripheral awareness of the display, Space B, characterized by focal awareness, in which visitors are engaged in activities associated with the display and Space C, characterized by direct interaction with the system [BR03]. These spaces can be understood as broadly corresponding to stages in Wouters' model. Space A encompasses forms of engagement such as the passer-by and bystander, Space B corresponds to the audience member, and Space C corresponds to the participant. Importantly, Brignull's model suggests that visitors must cross a threshold in order to move from one space to another. This provides a useful conceptualization for the present research, which focuses specifically on the threshold leading to direct interaction. This research is focused on the transition from audience member to participant: the point at which a visitor moves from being aware of an interactive installation to actively interacting with it.

Throughout this paper, "engagement" is used as the broader, umbrella term encompassing any form of visitor involvement with an installation, including passive observation, consistent with its use across all three source studies. "Participation," by contrast, is reserved specifically for the audience member → participant transition that is the focus of this paper. Both models were developed in the context of the interactive installations described in the Related Work Section. The study of Hornecker and Stifter is particularly relevant because it considers visitor behavior across different types of exhibits within the same exhibition context, providing insight into how the characteristics of an installation can influence whether and how visitors engage with it. In addition, the context of the installations - an exhibition in a curated space - is the closest match to the research questions at hand. Several recurring patterns emerge from Hornecker and Stifter's [HS06] study of a museum exhibition, Wouters et al.'s [WDH*16] Honeypot Model, and Brignull and Rogers' [BR03] analysis of public context: (3.1) the importance of legibility in the first seconds, (3.2) the honeypot effect of seeing others interact, and (3.3) the social risk of failing. Each of these factors is discussed below, concluding in a set of design implications for lowering the participation threshold in curated interactive installations.

3.1. First Seconds / Legibility

Before an installation can be approached or observed, it must first be noticed at all, which is why legibility is included here as a structural precondition for every later threshold, even though it is the least consistently evidenced factor in the framework. Hornecker and Stifter [HS06] offer the most direct claim: that "the first ten seconds need to provide an incentive to continue," though no supporting data accompanies this figure, and neither other study confirmed it. Brignull and Rogers [BR03] instead report a purely spatial measure: the Opinionizer's screen was "legible from approximately 5 meters away, while the screen itself was visible from anywhere in the room", describing whether the display could be seen, not whether visitors understood it at a glance. Notably, this visibility alone did not translate into early engagement: at the start of the party, "few people came forward to try out the Opinionizer," despite the display already being fully legible, suggesting legibility may be necessary but insufficient for approach (taken up in Section 3.2). Wouters et al. [WDH*16] frame initial awareness differently again, relying on triggers like signage, pitchmen, audiovisual output that were needed specifically because "the spatial configuration of Encounters prevented the interaction activity to be noticed from surrounding streets." Overall, these statements cannot be traced back to a common, measurable concept of legibility: Hornecker's claim remains an unconfirmed assumption, Brignull's observation pertains more to mere visibility than to comprehensibility, and Wouters' case shows that legibility from a distance fails

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entirely. This paper considers legibility for structural reasons but points out that, unlike the factors in Sections 3.2 and 3.3, it is not explicitly attested in the three sources themselves.

3.2. Honeypot Effect

The term "honeypot effect" was first coined by Brignull and Rogers [BR03] in their study of the Opinionizer. They describe this as a "sociable buzz," where the growing number of people around an installation attracts further bystanders and encourages them to participate. This effect is driven not only by the presence of others, but also by the conversations and comments they share about the installation [BR03].

Wouters et al. [WDH*16] build on this concept, identifying what they term a "honeypot sweet spot": while the number of people simultaneously interacting initially increases the motivation of others to join, participation declines once a certain threshold is exceeded. This complicates the earlier assumption that the honeypot effect is purely self-reinforcing, revealing instead a natural capping mechanism shaped by the installation's physical and spatial constraints, which was in their case study, the installation's dimensions that allowed only a fixed number of people to stand together at once [WDH*16]. This finding resonates with Brignull and Rogers' [BR03] earlier observation of "bottlenecks" in the flow of people around a display, which likewise shape how many people can physically engage at once.

Wouters et al. [WDH*16] quantify this threshold precisely: across their observation period, participant numbers per dance zone stabilized at an average of 3.1 (SD=0.8), a figure they attribute to a combination of physical constraints and a loss of individual recognizability on the shared screen once the zone became crowded.

Wouters et al. [WDH*16] further argue that their "activation loop" expands the traditional understanding of the honeypot effect as "solely relying on watching others," since it is "one of its key elements", consisting of the exchange of information between new participants and those who have already disengaged, and between bystanders and audience members learning social norms by observing participants and actors. As such, they conclude the effect "relies on communication between the different user roles," not observation alone – similar to Brignull's observations.

Taken together, these findings suggest that the honeypot effect is neither purely observational nor infinitely self-reinforcing: it depends on prior legibility to draw initial attention (see Section 3.1), operates within a bounded capacity shaped by spatial and hardware constraints, and is sustained less by passive watching than by active exchange between user roles.

3.3. Social Embarrassment

The Honeypot Effect suggests that social validation can play a role in the decision to participate. This can also be seen in the interviews conducted by Brignull and Rogers [BR03], which showed that social embarrassment can act as a barrier to interaction. Participants who chose not to contribute identified the possibility of feeling embarrassed as an important reason for not participating. Even among those who did participate, many reported feeling uncomfortable or self-conscious because they were aware that other people were watching them.

Brignull and Rogers also explored the use of mobile phones as an alternative input method. Allowing participants to submit their responses anonymously reduced the pressure associated with being observed and made it easier for them to formulate and submit a response. The fact that anonymity could reduce this barrier provides further evidence that social embarrassment is a relevant factor in the decision to participate. These findings suggest that fear of embarrassment can affect whether visitors interact with an installation. This raises the question of what causes this fear and which individual factors contribute to it.

3.3.1. Possibility of Failure

One potential source of embarrassment is the possibility of failure itself: if an installation permits no wrong outcome, there is structurally nothing to be embarrassed about - at least not on the level of task performance. The installations examined varied considerably in whether a failure state existed at all. In the TV Newsroom studio at medien.welten [HS06], visitors were guided through a simulated news broadcast, either reading a provided script or improvising freely, with no evaluative consequence attached to either choice, there was no wrong way to perform. Encounters [WDH*16] followed a similar logic: since the installation translated any movement into visual and audio output, visitors could not act "incorrectly" from a participant's perspective. Failure existed only as a technical possibility. The Abacus exhibit, an exhibit at medien.welten, by contrast, involved a genuine correct or incorrect calculation, introducing a real failure state. The Opinionizer [BR03] occupies a middle position: typing an opinion technically ruled out a "wrong" answer, yet whatever was written was exposed to the room in real time. Taken together, these cases suggest that failure-possibility alone is an insufficient predictor of perceived risk. Encounter and Opinionizer show that removing the possibility of a "wrong answer" does not necessarily prevent embarrassment. This suggests that visibility (Section 3.3.2) may play a greater role than failure itself.

3.3.2. Visibility of Failure

While the introduction established that visibility can act as a barrier to participation, the cases examined here show that this barrier operates very differently depending on the installation. The Opinionizer's exposure, as discussed above, correlated directly with reported embarrassment and avoidance. The Newsroom studio however, complicates this picture: despite being highly visible to the entire room, "the public visibility of both the videos and the installation did not seem to inhibit most visitors." [HS06]. Here, visibility correlated with popularity: the Newsroom was the single most-used exhibit in the museum, with visitors returning repeatedly and appropriating it creatively rather than avoiding it out of self-consciousness.

This paper suggests two possible explanations for this exception. First, the Newsroom's script drew on the familiar, low-stakes genre of reading news aloud, rather than requiring the spontaneous self-expression that made the Opinionizer risky. Second, and more

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speculatively, its predominantly group-based use may have allowed individuals to diffuse exposure across the group rather than facing the audience alone. This possibility is not directly addressed in Hornecker and Stifter's data, but consistent with their observation that families and school classes used the exhibit collaboratively.

The Abacus offers a further contrast: although failure was genuinely possible, it occurred within small, self-formed groups who supported each other by reading instructions aloud and discussing steps together. This shielding appears to have converted potential failure into shared achievement with visitors observed exclaiming "Hurray – I made it!" upon completion, even though many in fact abandoned the calculation partway through [HS06].

Encounters [WDH*16] appears to separate exposure from the possibility of failure. Although the installation did not involve any task that could be performed incorrectly, several audience members still identified fear of embarrassment as a reason for avoiding the dance zones or leaving them quickly. This suggests that being observed while performing an action, regardless of whether that action can be considered "wrong" can create discomfort and discourage participation. In this sense, visibility may produce a similar form of avoidance to the fear of failure observed in Opinionizer.

These four cases suggest that exposure, not failure-possibility, is the more consistent predictor of perceived social risk: Newsroom (exposed, no failure, group-shielded genre) thrived; Opinionizer (exposed, failure possible, individually authored) generated measurable avoidance; Abacus (failure possible, group-shielded) suggests that the risk of failing may be perceived as lower when a task is completed collectively; and Encounters (exposed, no failure) shows that exposure alone can suppress participation even without any fail state. This pattern returns when considering what kind of exposure (cognitive-task vs. personal-expressive) may matter most.

3.3.3. Anticipating the Risk

Beyond whether failure was possible (Section 3.3.1) and visible (Section 3.3.2), visitors' decisions to cross the participation threshold depended on whether they could anticipate what participation would involve before committing. Brignull and Rogers [BR03] identify uncertainty as a primary barrier to participation at the Opinionizer. When visitors do not know what to expect, they may be more likely to rely on their existing reluctance to engage. The authors therefore suggest that participation can be encouraged through vicarious learning, where visitors first observe others using the installation before joining themselves. A simple and easily understood interface further reduces the uncertainty involved in initiating interaction. From this, they highlight five questions visitors need answered before committing: how long an interaction takes, what they will get out of it, what steps are involved, whether it will be comfortable, and whether there is a quick, graceful way to withdraw. Their design implication is to keep interaction lightweight and easy to understand. Visitors should be able to learn by observing others, while mistakes should have minimal social consequences.

The Newsroom studio [HS06] illustrates both mechanisms at once. Its high visibility from across the room meant visitors could observe others performing before attempting it themselves, which illustrates Brignull's vicarious learning in practice. At the same time, unlike Encounters or the Opinionizer, the Newsroom also provided a concrete, given script (a news text on a line-by-line display), directly specifying the steps involved. The combination observability plus an explicit procedure may help explain why the Newsroom was both highly used and frequently appropriated creatively rather than avoided.

Wouters et al. [WDH*16], however, complicate a purely "more information is better" interpretation. In their case, ambiguous signage was more effective than explicit descriptions at initially attracting passers-by to Encounters. More detailed information from on-site pitchmen mainly benefited visitors who were already interested in attending. These findings concern only the early transition from passer-by to bystander, rather than the participation threshold itself. Wouters et al. therefore suggest preserving some ambiguity at this initial "rousing" stage, while also noting that visitors used vicarious peer learning, such as observing others and overhearing conversations, to reduce uncertainty before participating.

Overall, these results suggest that anticipatory clarity operates differently depending on which threshold is at stake: vague, intriguing information is more effective at drawing bystanders toward an installation, while concrete anticipatory clarity about duration, steps, comfort, and exit becomes more important closer to the participation threshold itself, where Brignull's five questions and the Newsroom's script both operate. Ambiguity and clarity are therefore not competing strategies but tools suited to different stages of the same trajectory.

4. Design Implications for Interactive Installations

To summarize the above analysis, several specific strategies for lowering the barrier to participation emerge.

Ensure recognizability from a distance. As discussed in Section 3.1, legibility remains the least well-evidenced factor in this synthesis, yet Brignull and Rogers' [BR03] own design choices point toward a practical minimum: deliberate sizing and positioning for distance legibility. Wouters et al.'s [WDH*16] experience with Encounters offers a cautionary counterpoint, where poor street-level visibility had to be compensated for after the fact through external signage and pitchmen. This suggests legibility is best treated as a first-order spatial design decision, not something upgraded once a site is fixed.

Support observation without forcing early commitment. The honeypot effect (Section 3.2) depends in part on bystanders being able to watch others for as long as they need before approaching, which depends on dedicated, comfortable space for spectating as seen in both the Newsroom's balustrade area and Wouters et al.'s [WDH*16] explicit recommendation to accommodate passive spectating. At the same time, Section 3.2 showed this cannot scale indefinitely: curators should dimension interaction zones with an upper bound in mind, not only a lower threshold to attract attention.

Shield potential failure within groups. Where an installation involves a genuine failure state (Section 3.3.1), the evidence favors designs that allow failure to be absorbed collectively rather than ex-

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posed individually. The Abacus's group-scaffolding pattern (Section 3.3.2) was a structural feature of its physical design: enough shared space for several people to stand and view the task together and not an incidental behavior. It stands in direct contrast to the Opinionizer's individually broadcast exposure.

Provide lightweight anticipatory cues. Section 3.3.3 showed that visitors need answers to a small set of concrete questions before committing, and that combining observability with an explicit procedure (as in the Newsroom) supports this. Critically, however, Wouters et al.'s [WDH*16] findings caution against applying this principle too early: ambiguity at the rousing stage was more effective at drawing passers-by than explicit description. The design implication is therefore staged rather than uniform: withhold procedural detail to generate curiosity from a distance but supply it clearly once a visitor is close enough to be weighing actual participation.

Together, these strategies reframe hesitation not as a trait of reluctant individual visitors, but as a predictable response to spatial and social conditions that installation design can actively shape.

5. Discussion

The sweet spot as a design lever. The preceding analysis has shown that the honeypot effect is a genuinely useful mechanism for drawing bystanders toward participation but not an unlimited one: Section 3.2 identified a "sweet spot" beyond which additional crowding reduces motivation to join. Crucially, this sweet spot is not a fixed psychological ceiling but a variable shaped by spatial and hardware choices. Wouters et al. [WDH*16] trace it to a concrete physical constraint, the archway's width limiting how many people could fit under the sensor at once. Reframed this way, crowd-size limitations become a design lever rather than an unavoidable constraint.

Beyond failure: embarrassment without a fail state. It is worth noting explicitly that Section 3.3.1 addresses failure-related social risk specifically, not social risk in general. Both Encounters and the Opinionizer suggest that a substantial part of visitors' embarrassment may not be about failure at all, but about self-exposure: dancing is an exposing, embodied act regardless of outcome, and Brignull and Rogers' [BR03] interview data points less toward fear of a technically "wrong" answer than toward fear of exposing one's own opinion or taste to public judgment. This is consistent with the framework rather than a contradiction of it: Sections 3.3.1 - 3.3.3 were designed to unpack the fear of failing, and both cases suggest that social embarrassment in public installations is not reducible to that fear alone.

Limitations. First, the three studies span thirteen years (2003–2016) and markedly different technological contexts, which complicates direct comparison. Hornecker and Stifter's [HS06] observation that visitors, particularly older ones, avoided screen-based exhibits while readily engaging with hands-on installations may partly reflect a level of digital unfamiliarity specific to its time; given today's higher baseline of screen literacy, it is plausible this finding would not replicate in the same form now, though the underlying mechanism (avoidance of unfamiliar interfaces) may still hold.

Second, all three source studies rely on qualitative observation and interviews rather than controlled experimentation. This reflects the nature of the phenomenon under investigation since social behavior around public installations is difficult to isolate under lab conditions without stripping away the very dynamics this paper's framework depends on rather than a flaw specific to the studies chosen. The consequence for this synthesis is that its conclusions rely on a relatively small number of specific installations and interview samples (e.g. 23 and 30 interviewees in the Brignull and Rogers study, 30 in Hornecker and Stifter) rather than a statistically representative range. The framework is therefore a synthesis of similar patterns found across different qualitative studies, rather than a causally or statistically validated model.

6. Conclusion

This paper began with a simple observation: many visitors to interactive installations remain observers, never crossing into participation. What this synthesis has shown is that this gap is not a fixed property of individual visitors, but something installations themselves create and can therefore also close. Legibility, the pull of watching others, and the social risk of being seen to fail are not incidental side effects of an installation's design; they are the design. Treating participation as an engineering problem rather than a matter of visitor disposition reframes what curators and designers are actually responsible for: not simply building something interactive, but building the conditions under which people feel safe enough to try. This shifts attention from the characteristics of individual visitors to the demands created by the installation and its surrounding environment.

7. Acknowledgements

I acknowledge the use of "Claude Sonnet 5" (available at https://claude.ai/) for phrasing/editing sentences and for locating specific passages within sources.

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